

DIGITAL FORENSIC INVESTIGATION REPORT

Insider Threat - Data Exfiltration Analysis

Case Number: CASE-FORENSIC-001

Subject: John Smith (EMP-2024-0892)

Date: April 05, 2026

Prepared By: NFLIP Automated Investigation System

CONFIDENTIAL - LAW ENFORCEMENT SENSITIVE

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1. Executive Summary

CRITICAL FINDING: This investigation has conclusively established unauthorized data exfiltration by John Smith (Employee ID: EMP-2024-0892) from corporate workstation DESKTOP-JXK92M through multiple channels over a 13-day period.

1.1 Investigation Overview

This report presents the findings of a comprehensive digital forensic investigation into suspected unauthorized data exfiltration by an employee. The investigation was initiated following anomalous behavior detection by the Security Operations Center (SOC) on March 19, 2026.

The investigation analyzed multiple digital evidence sources including Windows Event Logs, NTFS file system artifacts, Exchange Server message tracking logs, network traffic captures, and Bluetooth communication logs. Advanced analytical techniques including machine learning-based anomaly detection and temporal correlation analysis were employed.

1.2 Key Metrics

3.9 GB

Total Data Exfiltrated

85

Confidential Files Stolen

4

Exfiltration Channels Used

98%

Overall Confidence Score

1.3 Summary of Findings

Channel	Data Volume	File Count	Confidence	Status
USB Transfer	2.3 GB	47 files	95%	CONFIRMED
Email	290.3 MB	15 emails	88%	CONFIRMED
Bluetooth	890 MB	23 files	82%	CONFIRMED
Network (Cloud)	450 MB	Unknown	79%	CONFIRMED

Executive Summary (continued)

1.1 Detailed Analysis Section 1

This section provides additional context for the investigation findings. The forensic examination utilized state-of-the-art tools and methodologies to ensure complete and accurate analysis of all digital evidence.

Key aspects of the investigation methodology included forensic imaging with SHA-256 verification, chain of custody documentation, multi-tool analysis approach, and Bayesian hypothesis testing with ODNI ICD 203 confidence classification.

The investigation team analyzed over 20,000 log entries from multiple sources including Windows Security Event Logs, Sysmon telemetry, NTFS USN Journal, Exchange Server message tracking, network flow data, and Bluetooth protocol captures.

Evidence preservation followed strict forensic protocols with write-blocking during acquisition, cryptographic hash verification at each analysis stage, and detailed documentation of all examiner actions.

Evidence Timeline for Section 1

Time	Event	Source	Significance
14:32:15	USB Device Connected	Security Event 6416	Primary exfiltration vector
14:35:22	File Copy Started	USN Journal	Mass transfer initiated
16:45:00	Transfer Complete	NTFS MFT	47 files copied

Executive Summary (continued)

1.2 Detailed Analysis Section 2

This section provides additional context for the investigation findings. The forensic examination utilized state-of-the-art tools and methodologies to ensure complete and accurate analysis of all digital evidence.

Key aspects of the investigation methodology included forensic imaging with SHA-256 verification, chain of custody documentation, multi-tool analysis approach, and Bayesian hypothesis testing with ODNI ICD 203 confidence classification.

The investigation team analyzed over 20,000 log entries from multiple sources including Windows Security Event Logs, Sysmon telemetry, NTFS USN Journal, Exchange Server message tracking, network flow data, and Bluetooth protocol captures.

Evidence preservation followed strict forensic protocols with write-blocking during acquisition, cryptographic hash verification at each analysis stage, and detailed documentation of all examiner actions.

Evidence Timeline for Section 2

Time	Event	Source	Significance
14:32:15	USB Device Connected	Security Event 6416	Primary exfiltration vector
14:35:22	File Copy Started	USN Journal	Mass transfer initiated
16:45:00	Transfer Complete	NTFS MFT	47 files copied

Executive Summary (continued)

1.3 Detailed Analysis Section 3

This section provides additional context for the investigation findings. The forensic examination utilized state-of-the-art tools and methodologies to ensure complete and accurate analysis of all digital evidence.

Key aspects of the investigation methodology included forensic imaging with SHA-256 verification, chain of custody documentation, multi-tool analysis approach, and Bayesian hypothesis testing with ODNI ICD 203 confidence classification.

The investigation team analyzed over 20,000 log entries from multiple sources including Windows Security Event Logs, Sysmon telemetry, NTFS USN Journal, Exchange Server message tracking, network flow data, and Bluetooth protocol captures.

Evidence preservation followed strict forensic protocols with write-blocking during acquisition, cryptographic hash verification at each analysis stage, and detailed documentation of all examiner actions.

Evidence Timeline for Section 3

Time	Event	Source	Significance
14:32:15	USB Device Connected	Security Event 6416	Primary exfiltration vector
14:35:22	File Copy Started	USN Journal	Mass transfer initiated
16:45:00	Transfer Complete	NTFS MFT	47 files copied

Executive Summary (continued)

1.4 Detailed Analysis Section 4

This section provides additional context for the investigation findings. The forensic examination utilized state-of-the-art tools and methodologies to ensure complete and accurate analysis of all digital evidence.

Key aspects of the investigation methodology included forensic imaging with SHA-256 verification, chain of custody documentation, multi-tool analysis approach, and Bayesian hypothesis testing with ODNI ICD 203 confidence classification.

The investigation team analyzed over 20,000 log entries from multiple sources including Windows Security Event Logs, Sysmon telemetry, NTFS USN Journal, Exchange Server message tracking, network flow data, and Bluetooth protocol captures.

Evidence preservation followed strict forensic protocols with write-blocking during acquisition, cryptographic hash verification at each analysis stage, and detailed documentation of all examiner actions.

Evidence Timeline for Section 4

Time	Event	Source	Significance
14:32:15	USB Device Connected	Security Event 6416	Primary exfiltration vector
14:35:22	File Copy Started	USN Journal	Mass transfer initiated
16:45:00	Transfer Complete	NTFS MFT	47 files copied

Executive Summary (continued)

1.5 Detailed Analysis Section 5

This section provides additional context for the investigation findings. The forensic examination utilized state-of-the-art tools and methodologies to ensure complete and accurate analysis of all digital evidence.

Key aspects of the investigation methodology included forensic imaging with SHA-256 verification, chain of custody documentation, multi-tool analysis approach, and Bayesian hypothesis testing with ODNI ICD 203 confidence classification.

The investigation team analyzed over 20,000 log entries from multiple sources including Windows Security Event Logs, Sysmon telemetry, NTFS USN Journal, Exchange Server message tracking, network flow data, and Bluetooth protocol captures.

Evidence preservation followed strict forensic protocols with write-blocking during acquisition, cryptographic hash verification at each analysis stage, and detailed documentation of all examiner actions.

Evidence Timeline for Section 5

Time	Event	Source	Significance
14:32:15	USB Device Connected	Security Event 6416	Primary exfiltration vector
14:35:22	File Copy Started	USN Journal	Mass transfer initiated
16:45:00	Transfer Complete	NTFS MFT	47 files copied

2. Investigation Methodology

2.1 Forensic Analysis Phase 1

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 1: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 1

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.2 Forensic Analysis Phase 2

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 2: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 2

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.3 Forensic Analysis Phase 3

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Analysis Phase 3: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 3

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
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X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.4 Forensic Analysis Phase 4

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 4: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 4

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.5 Forensic Analysis Phase 5

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 5: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 5

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.6 Forensic Analysis Phase 6

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 6: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 6

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

2. Investigation Methodology

2.7 Forensic Analysis Phase 7

The investigation employed a systematic approach following NIST SP 800-86 guidelines for digital forensic analysis. Each phase of the investigation was documented with detailed notes and cryptographic verification of evidence integrity.

Analysis Phase 7: This phase focused on extraction and analysis of specific forensic artifacts including registry keys, file system metadata, and application-specific data structures.

Tools Used in Phase 7

Tool	Version	Purpose
FTK Imager	4.7.1	Forensic acquisition
Autopsy	4.21.0	Artifact analysis
X-Ways	21.1	Advanced recovery
NFLIP	2.4	AI-assisted analysis

The analysis identified multiple indicators of compromise (IOCs) consistent with insider threat behavior patterns. These IOCs were correlated across data sources to build a comprehensive picture of the exfiltration campaign.

3. Evidence Collection

3.1 Evidence Source Analysis 1

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-015	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-016	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-017	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.2 Evidence Source Analysis 2

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-016	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-017	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-018	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.3 Evidence Source Analysis 3

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-017	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-018	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-019	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.4 Evidence Source Analysis 4

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-018	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-019	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-020	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.5 Evidence Source Analysis 5

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-019	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-020	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-021	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.6 Evidence Source Analysis 6

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-020	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-021	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-022	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

3. Evidence Collection

3.7 Evidence Source Analysis 7

This section documents the collection and preservation of digital evidence from multiple sources. All evidence was acquired using forensically sound methods with write-blocking and cryptographic hash verification.

Evidence Items Collected

ID	Type	Source	Size	Hash (SHA-256)
EVD-021	Disk Image	DESKTOP-JXK92M	256 GB	a7f3c9d2...
EVD-022	Memory	RAM Capture	32 GB	c4d2e8f1...
EVD-023	Logs	Event Logs	1.2 GB	b8e4f1c7...

Chain of Custody: All evidence was collected, handled, and stored in accordance with organizational chain of custody procedures. Dual-examiner verification was employed for all critical evidence items.

Evidence integrity was verified at each stage of analysis using SHA-256 cryptographic hashes. No evidence tampering or modification was detected throughout the investigation process.

4. USB Exfiltration Analysis

4.1 USB Forensic Details 1

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 1)

#	Filename	Size	Classification
4	Document_4.xlsx	45.2 MB	CONFIDENTIAL
5	Report_5.pdf	23.1 MB	INTERNAL
6	Data_6.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.2 USB Forensic Details 2

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 2)

#	Filename	Size	Classification
7	Document_7.xlsx	45.2 MB	CONFIDENTIAL
8	Report_8.pdf	23.1 MB	INTERNAL
9	Data_9.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.3 USB Forensic Details 3

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 3)

#	Filename	Size	Classification
10	Document_10.xlsx	45.2 MB	CONFIDENTIAL
11	Report_11.pdf	23.1 MB	INTERNAL
12	Data_12.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.4 USB Forensic Details 4

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 4)

#	Filename	Size	Classification
13	Document_13.xlsx	45.2 MB	CONFIDENTIAL
14	Report_14.pdf	23.1 MB	INTERNAL
15	Data_15.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.5 USB Forensic Details 5

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 5)

#	Filename	Size	Classification
16	Document_16.xlsx	45.2 MB	CONFIDENTIAL
17	Report_17.pdf	23.1 MB	INTERNAL
18	Data_18.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.6 USB Forensic Details 6

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 6)

#	Filename	Size	Classification
19	Document_19.xlsx	45.2 MB	CONFIDENTIAL
20	Report_20.pdf	23.1 MB	INTERNAL
21	Data_21.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.7 USB Forensic Details 7

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 7)

#	Filename	Size	Classification
22	Document_22.xlsx	45.2 MB	CONFIDENTIAL
23	Report_23.pdf	23.1 MB	INTERNAL
24	Data_24.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.8 USB Forensic Details 8

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 8)

#	Filename	Size	Classification
25	Document_25.xlsx	45.2 MB	CONFIDENTIAL
26	Report_26.pdf	23.1 MB	INTERNAL
27	Data_27.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.9 USB Forensic Details 9

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 9)

#	Filename	Size	Classification
28	Document_28.xlsx	45.2 MB	CONFIDENTIAL
29	Report_29.pdf	23.1 MB	INTERNAL
30	Data_30.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.10 USB Forensic Details 10

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 10)

#	Filename	Size	Classification
31	Document_31.xlsx	45.2 MB	CONFIDENTIAL
32	Report_32.pdf	23.1 MB	INTERNAL
33	Data_33.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.11 USB Forensic Details 11

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 11)

#	Filename	Size	Classification
34	Document_34.xlsx	45.2 MB	CONFIDENTIAL
35	Report_35.pdf	23.1 MB	INTERNAL
36	Data_36.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.12 USB Forensic Details 12

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 12)

#	Filename	Size	Classification
37	Document_37.xlsx	45.2 MB	CONFIDENTIAL
38	Report_38.pdf	23.1 MB	INTERNAL
39	Data_39.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

4. USB Exfiltration Analysis

4.13 USB Forensic Details 13

Analysis of USB device connection and file transfer activity revealed systematic exfiltration of confidential data. Windows registry artifacts, NTFS journal entries, and prefetch files provided comprehensive evidence of the transfer activity.

USB Device Information

Device: Samsung Galaxy S23
Serial: RF8N30ABCDE
VID/PID: 04E8/6860
First Connection: 2026-03-20 14:32:15 UTC

Files Transferred (Subset 13)

#	Filename	Size	Classification
40	Document_40.xlsx	45.2 MB	CONFIDENTIAL
41	Report_41.pdf	23.1 MB	INTERNAL
42	Data_42.csv	89.4 MB	CONFIDENTIAL

Finding: USB data exfiltration confirmed with 95% confidence based on registry, journal, and prefetch evidence.

5. Email Exfiltration Analysis

5.1 Email Forensic Details 1

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 11	Meeting Notes	Report.xlsx	12.4 MB
Mar 12	FYI	Data.csv	8.9 MB
Mar 13	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.2 Email Forensic Details 2

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 12	Meeting Notes	Report.xlsx	12.4 MB
Mar 13	FYI	Data.csv	8.9 MB
Mar 14	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.3 Email Forensic Details 3

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 13	Meeting Notes	Report.xlsx	12.4 MB
Mar 14	FYI	Data.csv	8.9 MB
Mar 15	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.4 Email Forensic Details 4

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 14	Meeting Notes	Report.xlsx	12.4 MB
Mar 15	FYI	Data.csv	8.9 MB
Mar 16	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.5 Email Forensic Details 5

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 15	Meeting Notes	Report.xlsx	12.4 MB
Mar 16	FYI	Data.csv	8.9 MB
Mar 17	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.6 Email Forensic Details 6

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 16	Meeting Notes	Report.xlsx	12.4 MB
Mar 17	FYI	Data.csv	8.9 MB
Mar 18	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.7 Email Forensic Details 7

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 17	Meeting Notes	Report.xlsx	12.4 MB
Mar 18	FYI	Data.csv	8.9 MB
Mar 19	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.8 Email Forensic Details 8

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 18	Meeting Notes	Report.xlsx	12.4 MB
Mar 19	FYI	Data.csv	8.9 MB
Mar 20	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.9 Email Forensic Details 9

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 19	Meeting Notes	Report.xlsx	12.4 MB
Mar 20	FYI	Data.csv	8.9 MB
Mar 21	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

5. Email Exfiltration Analysis

5.10 Email Forensic Details 10

Exchange Server message tracking logs revealed systematic transmission of confidential attachments to personal email addresses. SMTP headers and Outlook PST analysis corroborated the exfiltration activity.

Email Summary

Date	Subject	Attachments	Size
Mar 20	Meeting Notes	Report.xlsx	12.4 MB
Mar 21	FYI	Data.csv	8.9 MB
Mar 22	Updates	Summary.pdf	15.7 MB

DLP bypass techniques identified include ZIP compression of attachments, obfuscated subject lines, and throttled transmission to avoid volume-based triggers.

Finding: Email exfiltration confirmed with 88% confidence. 15 emails containing 290.3MB sent to personal Gmail.

6. Network Exfiltration Analysis

6.1 Network Traffic Analysis 1

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.2 Network Traffic Analysis 2

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.3 Network Traffic Analysis 3

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.4 Network Traffic Analysis 4

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.5 Network Traffic Analysis 5

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.6 Network Traffic Analysis 6

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.7 Network Traffic Analysis 7

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.8 Network Traffic Analysis 8

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.9 Network Traffic Analysis 9

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

6. Network Exfiltration Analysis

6.10 Network Traffic Analysis 10

Network traffic analysis revealed significant outbound data transfers to cloud storage services and VPN tunnels. Firewall logs, proxy records, and DNS query analysis provided comprehensive visibility into the network-based exfiltration channels.

Traffic Summary

Service	Destination	Sessions	Data
Dropbox	203.45.67.89	47	157 MB
Google Drive	142.250.185.78	23	89 MB
NordVPN	104.18.32.46	8	524 MB

VPN Tunnel: 3-hour connection to NordVPN on March 20, 2026 with 524MB of encrypted traffic.

DNS query analysis showed 340% increase in cloud storage domain lookups during the investigation period.

7. Bluetooth Transfer Analysis

7.1 Bluetooth Protocol Analysis 1

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
3	Mar 15	55 min	8	234 MB
4	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.2 Bluetooth Protocol Analysis 2

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
5	Mar 15	55 min	8	234 MB
6	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.3 Bluetooth Protocol Analysis 3

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
7	Mar 15	55 min	8	234 MB
8	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.4 Bluetooth Protocol Analysis 4

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
9	Mar 15	55 min	8	234 MB
10	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.5 Bluetooth Protocol Analysis 5

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
11	Mar 15	55 min	8	234 MB
12	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.6 Bluetooth Protocol Analysis 6

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
13	Mar 15	55 min	8	234 MB
14	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

7. Bluetooth Transfer Analysis

7.7 Bluetooth Protocol Analysis 7

Windows Bluetooth stack logs confirmed device pairing and file transfer activity using OBEX File Transfer Profile. 12 transfer sessions were recorded over a 3-day period.

Transfer Sessions

Session	Date	Duration	Files	Size
15	Mar 15	55 min	8	234 MB
16	Mar 16	40 min	5	156 MB

Finding: Bluetooth transfer confirmed with 82% confidence. 23 files totaling 890MB transferred via OBEX.

8. Timeline Analysis

8.1 Unified Timeline Section 1

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.2 Unified Timeline Section 2

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.3 Unified Timeline Section 3

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.4 Unified Timeline Section 4

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.5 Unified Timeline Section 5

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.6 Unified Timeline Section 6

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.7 Unified Timeline Section 7

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

8. Timeline Analysis

8.8 Unified Timeline Section 8

Events were reconstructed from 20,499 log entries across all sources. 847 events were directly related to the investigation, spanning March 10-22, 2026.

847

Related Events

13

Days Duration

Activity Clusters

Cluster	Date Range	Channel	Events
1	Mar 10-12	Email	23
2	Mar 15-17	Bluetooth	89
3	Mar 18-19	Network	67
4	Mar 20	USB	127

9. Anomaly Detection

9.1 ML Analysis Section 1

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.2 ML Analysis Section 2

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.3 ML Analysis Section 3

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.4 ML Analysis Section 4

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.5 ML Analysis Section 5

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.6 ML Analysis Section 6

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.7 ML Analysis Section 7

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

9. Anomaly Detection

9.8 ML Analysis Section 8

Machine learning anomaly detection (Isolation Forest) identified significant behavioral deviations from established baselines. The subject's anomaly score of 0.89 exceeded the threshold of 0.65.

Baseline Deviations

Metric	Baseline	Actual	Deviation
Files accessed/day	45	198	+340%
USB connections	2/month	5/day	+7,400%
After-hours activity	4/month	8/13 days	+500%

SHAP Feature Importance

Feature	SHAP Value
file_access_count	+0.42
after_hours_activity	+0.31
sensitive_folder_access	+0.28

10. MITRE ATT&CK Mapping

10.1 Technique Analysis 1

The following MITRE ATT&CK techniques were identified in this investigation:

ID	Technique	Tactic	Evidence
T1052.001	Exfiltration over USB	Exfiltration	47 files transferred
T1048.002	Encrypted Channel	Exfiltration	VPN tunnel
T1567.002	Cloud Storage	Exfiltration	Dropbox, Drive
T1070.004	File Deletion	Defense Evasion	Anti-forensics

10. MITRE ATT&CK Mapping

10.2 Technique Analysis 2

The following MITRE ATT&CK techniques were identified in this investigation:

ID	Technique	Tactic	Evidence
T1052.001	Exfiltration over USB	Exfiltration	47 files transferred
T1048.002	Encrypted Channel	Exfiltration	VPN tunnel
T1567.002	Cloud Storage	Exfiltration	Dropbox, Drive
T1070.004	File Deletion	Defense Evasion	Anti-forensics

10. MITRE ATT&CK Mapping

10.3 Technique Analysis 3

The following MITRE ATT&CK techniques were identified in this investigation:

ID	Technique	Tactic	Evidence
T1052.001	Exfiltration over USB	Exfiltration	47 files transferred
T1048.002	Encrypted Channel	Exfiltration	VPN tunnel
T1567.002	Cloud Storage	Exfiltration	Dropbox, Drive
T1070.004	File Deletion	Defense Evasion	Anti-forensics

10. MITRE ATT&CK Mapping

10.4 Technique Analysis 4

The following MITRE ATT&CK techniques were identified in this investigation:

ID	Technique	Tactic	Evidence
T1052.001	Exfiltration over USB	Exfiltration	47 files transferred
T1048.002	Encrypted Channel	Exfiltration	VPN tunnel
T1567.002	Cloud Storage	Exfiltration	Dropbox, Drive
T1070.004	File Deletion	Defense Evasion	Anti-forensics

11. Hypothesis Evaluation

11.1 Evaluation Section 1

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

11. Hypothesis Evaluation

11.2 Evaluation Section 2

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

11. Hypothesis Evaluation

11.3 Evaluation Section 3

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

11. Hypothesis Evaluation

11.4 Evaluation Section 4

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

11. Hypothesis Evaluation

11.5 Evaluation Section 5

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

11. Hypothesis Evaluation

11.6 Evaluation Section 6

All hypotheses were evaluated using Bayesian probability assessment with ODNI ICD 203 confidence classification.

Hypothesis	Verdict	Confidence	ODNI Level
USB Exfiltration	CONFIRMED	95%	Very High
Email Exfiltration	CONFIRMED	88%	High
Bluetooth Transfer	CONFIRMED	82%	High
Network Exfiltration	CONFIRMED	79%	Moderate-High
Null Hypothesis	REJECTED	98%	Very High

FINAL VERDICT: Null hypothesis REJECTED with 98% confidence. Unauthorized multi-channel data exfiltration confirmed.

12. Conclusions and Recommendations

12.1 Recommendations Section 1

Based on the investigation findings, the following recommendations are made:

Immediate Actions

1. Preserve all evidence per chain of custody requirements
2. Forward findings to legal counsel for proceedings
3. Disable all access for the subject
4. Initiate data recovery procedures

Technical Recommendations

1. Implement DLP for ZIP archive inspection
2. Enable USB device whitelisting
3. Block personal cloud storage services
4. Disable Bluetooth file transfer

12. Conclusions and Recommendations

12.2 Recommendations Section 2

Based on the investigation findings, the following recommendations are made:

Immediate Actions

1. Preserve all evidence per chain of custody requirements
2. Forward findings to legal counsel for proceedings
3. Disable all access for the subject
4. Initiate data recovery procedures

Technical Recommendations

1. Implement DLP for ZIP archive inspection
2. Enable USB device whitelisting
3. Block personal cloud storage services
4. Disable Bluetooth file transfer

12. Conclusions and Recommendations

12.3 Recommendations Section 3

Based on the investigation findings, the following recommendations are made:

Immediate Actions

1. Preserve all evidence per chain of custody requirements
2. Forward findings to legal counsel for proceedings
3. Disable all access for the subject
4. Initiate data recovery procedures

Technical Recommendations

1. Implement DLP for ZIP archive inspection
2. Enable USB device whitelisting
3. Block personal cloud storage services
4. Disable Bluetooth file transfer

12. Conclusions and Recommendations

12.4 Recommendations Section 4

Based on the investigation findings, the following recommendations are made:

Immediate Actions

1. Preserve all evidence per chain of custody requirements
2. Forward findings to legal counsel for proceedings
3. Disable all access for the subject
4. Initiate data recovery procedures

Technical Recommendations

1. Implement DLP for ZIP archive inspection
2. Enable USB device whitelisting
3. Block personal cloud storage services
4. Disable Bluetooth file transfer

Appendix A

Supporting Documentation 1

This appendix contains supplementary technical information, evidence inventory details, and examiner certification documentation.

Evidence Hash Summary

Evidence ID	Type	SHA-256 Hash
EVD-092A	Registry	a7f3c9d2e8...b2e1f4a6
EVD-092B	Log File	c4d2e8f1a3...f9a3b7c5
EVD-092C	Memory	b8e4f1c7d2...c7d2a9e3

Verification: All hashes verified at analysis completion. Evidence integrity confirmed.

Appendix B

Supporting Documentation 2

This appendix contains supplementary technical information, evidence inventory details, and examiner certification documentation.

Evidence Hash Summary

Evidence ID	Type	SHA-256 Hash
EVD-093A	Registry	a7f3c9d2e8...b2e1f4a6
EVD-093B	Log File	c4d2e8f1a3...f9a3b7c5
EVD-093C	Memory	b8e4f1c7d2...c7d2a9e3

Verification: All hashes verified at analysis completion. Evidence integrity confirmed.

Appendix C

Supporting Documentation 3

This appendix contains supplementary technical information, evidence inventory details, and examiner certification documentation.

Evidence Hash Summary

Evidence ID	Type	SHA-256 Hash
EVD-094A	Registry	a7f3c9d2e8...b2e1f4a6
EVD-094B	Log File	c4d2e8f1a3...f9a3b7c5
EVD-094C	Memory	b8e4f1c7d2...c7d2a9e3

Verification: All hashes verified at analysis completion. Evidence integrity confirmed.

Appendix D

Supporting Documentation 4

This appendix contains supplementary technical information, evidence inventory details, and examiner certification documentation.

Evidence Hash Summary

Evidence ID	Type	SHA-256 Hash
EVD-095A	Registry	a7f3c9d2e8...b2e1f4a6
EVD-095B	Log File	c4d2e8f1a3...f9a3b7c5
EVD-095C	Memory	b8e4f1c7d2...c7d2a9e3

Verification: All hashes verified at analysis completion. Evidence integrity confirmed.

